25901185 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBCA201 - PROBABILITY DISTRIBUTIONS AND STATISTICAL INFERENCE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. The p.d.f of the random variable X is P(X = 1) = k, P(X = 2) = 2k, P(X = 3)= 3k. Find k. [A] / [CO1]
2. Mention one assumption of the chi-square test of independence. [U] / [CO4]
3. Find P(Z< 2.34) [A] / [CO2]
4. Write the degrees of freedom in a two-sample t-test (equal variance case)? If n1=12 and n2=15, find the degrees of freedom. [U] / [CO4]
5. What is the Sampling distribution of Variance? [U] / [CO3]
6. State the decision rule using the p-value approach. [U] / [CO4]
7. What is meant by standard normal distribution? [U] / [CO3]
8. Write the test statistic for testing a single population proportion. [K] / [CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

9. Define Expectation. Write the properties of expectation. [U] / [CO1]

10. Define Bernoulli distribution. Obtain its mean, variance and MGF. [U] / [CO2]

11. State the form of the F-distribution and explain its properties and uses. [U] / [CO3]

12. A random sample of 200 tins of coconut oil gives an average weight of 4.95 kg with a S.D of 0.21 kg. Do we accept the hypothesis of net weight of 5 kgs. per tin at 1% level of significance? [An] / [CO4]

13. If X is a Poisson variate such that P(X=1) = 2 P(X=2), find P(X =0). [An] / [CO2]

14. Explain briefly the fitting of Binomial distribution. [U] / [CO2]

15. Differentiate between independent sample t test and paired t test. [U] / [CO4]

16. Define Null and alternative hypothesis. Expalin with suitable example. [A] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

17. Define a random variable. Explain in detail the types of random variables by discussing their definitions, characteristics, differences, and real-life applications with examples. [An] / [CO1]

18. Explain Normal Distribution and Discuss its properties. [U] / [CO2]

19. a) Explain Statistic and parameter with example ? [K] / [CO3]
    b) Explain Standard Deviation and Standard Error ?
    c) What is the difference between Standard Deviation and Standard Error?

20. In an experiment on pea breeding mendel obtained the following frequencies of seeds 315 round and yellow, 101 wrinkled and yellow,108 round and green ,36 wrinkled and green . Theory predicts that the frequencies should be in the proportion 9:3:2:1.Examine the correspondence between theory and experiment. [A] / [CO4]

[15x2 = 30]